

58.5/65	90.0%	A+	11/12	32/33	15.5/20
TOTAL SCORE	PERCENTAGE	GRADE	SECTION A (MCQ)	SECTION B	SECTION C

All three sections present across 17 scanned pages (CamScanner), every page clearly legible at high zoom. Seemab answered 11 of the 14 Section B questions — skipping (vi), (x) and (xiv), an allowed choice — and 4 of the 5 Section C questions — skipping (iii), the cubical-tank problem, also allowed. **Integrity note:** Q3(v)(c) ("what type of quadrilateral is EFGH? Justify briefly") is the final sub-part of the last Section C question attempted. All four side lengths in part (b) immediately before it were computed correctly, but no answer for the classification appears anywhere on the scan — the page has substantial blank space left underneath, followed directly by the Section A MCQ answers on the very next scanned page. This pattern (trailing blank space, no missing-page signature) indicates a genuine non-attempt rather than an unscanned page, so it is marked 0/1 and flagged here rather than assumed. A second, independent read of every disputed answer was carried out before finalising this report: Q3(iv)(b)'s number line was confirmed to carry deliberate solution-point circles at -2 and 7 (full marks restored), while Q2(vii)(a) and Q2(ix) were found to drop or garble their final answer lines despite fully correct working underneath (half marks each, details below). The deduction log (6.5 marks lost) reconciles exactly with the headline 58.5/65. One answer-key discrepancy was also found and resolved in Seemab's favour (Q2(iv)(a), noted below).

Section A — Multiple Choice Questions (12 × 1 = 12 Marks)

MCQ answers listed on the final scanned page. 11 of 12 match the Version 1 key.

#	Topic	Key	Seemab	Result	Mark
1	Multiplicative inverse of 0	D	D	✓	1
2	Exponential form $5^{\frac{2}{3}}$: what the denominator (3) represents	B	B	✓	1
3	The mantissa of a logarithm is always	B	B	✓	1
4	Which logarithm is undefined: $\log_2(-4)$	C	C	✓	1
5	Reference position of 0.00532	A	D	✗	0
6	HCF of two polynomials unchanged when scaled by	A	A	✓	1
7	Factors of $x^2-11x+30$	A	A	✓	1
8	Square root of $49a^2-84ab+36b^2$	A	A	✓	1
9	Solution set of $ x-4 =7$	A	A	✓	1
10	Inequality direction, multiplied by $c=-2$	B	B	✓	1
11	Quadrant of point (3,−5)	D	D	✓	1
12	RS length, P midpoint of RS, PS=6	C	C	✓	1
Section A Total					11 / 12

One miss: Q5 — the reference position in 0.00532 lies immediately after the first significant digit (between the 5 and the 3); Seemab chose D ("at the decimal point") instead of A. Meets Seemab's 11-12/12 baseline, re-verified at 500 DPI zoom, high confidence.

Section B — Short Questions (11 of 14 attempted, 11 × 3 = 33 Marks)

Seemab answered 2(i), (ii), (iii), (iv), (v), (vii), (viii), (ix), (xi), (xii), (xiii) — skipping (vi), (x), (xiv). 9 of the 11 attempted questions scored full marks; two (vii, ix) lost 0.5 each to final-answer transcription slips after otherwise correct working.

Q2(i) — Unit 1: Scientific Notation Application

Distance = speed × time = $3.5 \times 12 = 42$ km = **4.2×10^1 km**. Correct throughout (3/3).

3/3

Q2(ii) — Unit 1: Real-Number Application (Temperature Scales)

$F = (9/5)(-89.2) + 32 = -160.56 + 32 = -128.56^\circ\text{F}$. $K = -89.2 + 273 = 183.8$ K. Both match the key exactly (3/3).

3/3

Q2(iii) — Unit 1: Properties of Equality & Inequality

(a) Transitive property of equality. Correct (1.5/1.5). (b) Multiplicative property of inequality. Correct (1.5/1.5).

3/3

Q2(iv) — Unit 2: Antilogarithm

(a) Characteristic=2, mantissa=.4324 → digits 2704+3=2707 → **antilog(2.4324) ≈ 270.7**. *Key discrepancy: the answer key states 270.8, but a precise calculation gives $10^{2.4324} = 270.645$, so Seemab's 270.7 is closer to the true value than the key's own stated answer — likely a table-rounding slip in the key. Full marks awarded per correct mathematics.* (b) Characteristic=1, mantissa=.5890 → digits 3882+0 → **antilog(1.5890) ≈ 38.82**, matches key exactly. Total (3/3).

3/3

Q2(v) — Unit 2: Domain of a Logarithm

$\log_2(7-x)$ checked for $x=0,1,6,7$: defined, defined, defined, undefined ($\log_2 0$ does not exist). All four correct (3/3).

3/3

Q2(vii) — Unit 4: Sum/Difference of Cubes & Squares

2.5/3

(a) $27+512x^3 = (3+8x)(9-24x+64x^2)$. **-0.5:** the sum-of-cubes working is correct throughout, but the final answer line drops the x^2 , written " $(3+8x)(9-24x+64)$ " instead of " $(3+8x)(9-24x+64x^2)$ " (1.0/1.5). (b) $a^6-64 = (a-2)(a+2)(a^2+2a+4)(a^2-2a+4)$. Correct (1.5/1.5). Total (2.5/3).

Q2(viii) — Unit 4: HCF by Factorization

3/3

$12x^2+x-1 = (4x-1)(3x+1)$; $15x^2+8x+1 = (5x+1)(3x+1)$; common factor $\Rightarrow$ **HCF = $(3x+1)$** . Correct throughout (3/3).

Q2(ix) — Unit 4: Square Root by Factorization

2.5/3

$25a^4-30a^3+9a^2 = a^2(5a-3)^2$, so the square root should conclude **$\pm(5a^2-3a)$** . **-0.5:** the square-root extraction is garbled at the end — the concluding lines " $\pm[a(5a-3)]^2$ " and " $\pm((5a^2-3a))^2$ " both carry a stray outer exponent 2, which is equal to the original quartic rather than its root; the clean $\pm(5a^2-3a)$ never actually appears on the page, even though it's clearly the intended answer from the factoring shown. Partial credit for the correct factoring (2.5/3).

Q2(xi) — Unit 5: Checking Solutions of Inequalities

3/3

(a) $5(3)-12=3>0 \Rightarrow$ true, $y=3$ satisfies. Correct (1.5/1.5). (b) $5-2(4)=-3 > -4(4)+5=-11 \Rightarrow$ true, $x=4$ satisfies. Correct (1.5/1.5).

Q2(xii) — Unit 5: Absolute Value Equation

3/3

$|5x-3| = |x+7|$: Case 1 $\rightarrow x=5/2$. Case 2 $\rightarrow x=-2/3$. Solution set $\{5/2, -2/3\}$, matches key exactly (3/3).

Q2(xiii) — Unit 7: Midpoint Formula (Decimals)

3/3

Midpoint of $(1.4,-1.5)$ and $(2.6,3.5) = ((1.4+2.6)/2, (-1.5+3.5)/2) =$ **$(2, 1)$** . Correct (3/3).

Section B Total

32 / 33

Section B held up well at 32/33 (97.0%) — 9 of the 11 attempted questions scored full marks, including the harder antilog and HCF-by-factorization items. The two half-mark losses (Q2(vii), Q2(ix)) were both final-line transcription slips after fully correct working, not conceptual gaps.

Section C — Long Questions (4 of 5 attempted, 4 × 5 = 20 Marks)

Seemab answered 3(i), (ii), (iv), (v) — skipping 3(iii), the cubical oil-tank problem.

Q3(i) — Unit 1: Trichotomy Property & Ratio Word Problem

3.5/5

(a) [2 marks] $-2\sqrt{3} \approx -2(1.732) = -3.464$, compared with $b=-5$: since $-3.464 > -5$, $a>b$, confirming the Trichotomy Property. Correct throughout (2/2).

(b) [3 marks] **-1.5:** misread the total penalty as **Rs. 6,000,000** instead of the **Rs. 600,000** stated in the question — a $10\times$ transcription error. The ratio method itself ($1+2+3=6$ parts, then multiply each partner's share) is completely correct and consistently applied, but every downstream Rupee figure ($1,000,000 / 2,000,000 / 3,000,000$, instead of the correct $100,000 / 200,000 / 300,000$) is $10\times$ too large as a result. Half credit awarded for the sound method despite the misread figure (1.5/3).

Q3(ii) — Unit 2: Laws of Logarithms & Digit Count

3.5/5

(a) [3 marks] $\log(28/45) = \log 4 + \log 7 - \log 9 - \log 5 = 2\log 2 + \log 7 - 2\log 3 - \log 5 = 2(0.3010) + 0.8451 - 2(0.4771) - 0.6990 = 1.4471 - 1.6532 =$ **-0.2061** , matches the key exactly (3/3).

(b) [2 marks] $\log(7^{40}) = 40\times\log 7 = 40\times 0.8451$. **-1.5:** computed this product as **34.04** instead of the correct **33.804** (a multiplication slip), which carried through to a wrong characteristic (34 instead of 33) and a final answer of **35 digits** instead of the correct **34 digits**. The method ($40 \log 7$, digits = characteristic+1) is entirely correct; only the arithmetic is wrong. Partial credit for the correct approach (0.5/2).

Q3(iv) — Unit 5: Compound Inequality & Absolute Value Equation

4.5/5

(a) [3 marks] Case 1: $5x+10<-35 \Rightarrow x<-9$, correct (1/1). Case 2: written as " $5x+10 < -5 \dots x < -3$ " then silently flipped to " $x > -3$ " with no justification — the final combined solution set **$\{x\in\mathbb{R} : x<-9 \text{ or } x>-3\}$** matches the key exactly, but the case-2 working contradicts its own written direction partway through. **-0.5:** case-2 mark 0.5/1. Combined-answer line correct (1/1). Subtotal (2.5/3).

(b) [2 marks] $|2y-5|=9 \Rightarrow y=7$ or $y=-2$, correctly solved (1.5/1.5 for the algebra). Number line: on deep-check, deliberate circles ARE plotted at the -2 and 7 ticks on the drawn scale (-9 to 8), correctly representing the solution set — full marks restored (0.5/0.5). Subtotal (2/2).

Q3(v) — Unit 7: Midpoint Quadrilateral (Varignon-type)

4/5

(a) [2 marks] $E=\text{midpoint}(AB)=(3,1)$, $F=\text{midpoint}(BC)=(7/2,4)$, $G=\text{midpoint}(CD)=(-1/2,4)$, $H=\text{midpoint}(DA)=(-1,1)$. All four correct (2/2).

(b) [2 marks] $EF=\sqrt{(1/4+9)}=\sqrt{37/2}$, $FG=\sqrt{16}=4$, $GH=\sqrt{(1/4+9)}=\sqrt{37/2}$, $HE=\sqrt{16}=4$. All four side lengths correct (2/2).

(c) [1 mark] **-1.0:** not attempted — despite $EF=GH$ and $FG=HE$ both correctly computed in part (b), which is exactly the evidence needed to conclude "EFGH is a parallelogram (both pairs of opposite sides equal)," no classification or justification sentence appears anywhere on the scan (0/1). See integrity note above: treated as a genuine non-attempt, not a missing page.

Section C Total

15.5 / 20

Final Score Breakdown

Section	Available	Earned	%	Notes
Section A — MCQ	12	11	91.7%	Lost 1: Q5 reference-position term (D instead of A)
Section B — Short Questions	33	32	97.0%	Lost 1.0: Q2(vii)(a) dropped x^2 in the cubes factorization (0.5), Q2(ix) stray exponent left on the final square-root line (0.5)
Section C — Long Questions	20	15.5	77.5%	Lost 4.5: misread penalty figure (1.5), $\log\times 40$ arithmetic slip (1.5), Q3(iv)(a) case-2 sign flip without justification (0.5), quadrilateral ID not attempted (1.0)

Deduction Log (all 6.5 marks lost)

Question	Sub-part	Issue	Lost
MCQ Q5	—	Reference position of 0.00532 answered D ("at the decimal point"); correct is A ("between 5 and 3", right after the first significant digit).	−1.0
Q2(vii)	(a) sum of cubes	Working correct throughout, but the final factorization line dropped the x^2 : wrote " $(3+8x)(9-24x+64)$ " instead of " $(3+8x)(9-24x+64x^2)$ ".	−0.5
Q2(ix)	square root	Concluding lines carried a stray outer exponent (" $\pm[a(5a-3)]^2$ ", " $\pm((5a^2-3a))^2$ ") equal to the original quartic rather than its root; the clean $\pm(5a^2-3a)$ was never written.	−0.5
Q3(i)	(b) ratio split	Misread the total penalty as Rs. 6,000,000 instead of the Rs. 600,000 given; ratio method (1:2:3, 6 parts) fully correct but every final amount is $10\times$ too large.	−1.5
Q3(ii)	(b) digit count	40×0.8451 computed as 34.04 instead of 33.804 (arithmetic slip); wrong characteristic (34 vs 33) gave 35 digits instead of the correct 34.	−1.5
Q3(iv)	(a) case-2 inequality	Case-2 working shown as " $5x+10 < -5 \dots x < -3$ " then silently flipped to " $x > -3$ " with no justification; the final combined solution set is correct but the written direction contradicts itself.	−0.5
Q3(v)	(c) classify EFGH	All four sides computed ($EF=GH$, $FG=HE$) but the quadrilateral type + justification was never written.	−1.0

Strengths in this attempt

- Ratio method solid despite the misread figure:** the 1:2:3 (6-part) ratio split in Q3(i)(b) was set up and applied correctly and consistently throughout — only the underlying Rupee figure was misread, never the method.
- Logarithm work strong across the paper:** $\log(28/45)$ via the laws of logarithms is fully correct, and Seemab's $\text{antilog}(2.4324) = 270.7$ is mathematically closer to the true value (270.645) than the answer key's own stated 270.8 — a genuine catch of a key error.
- Coordinate geometry (Varignon midpoint quadrilateral) mechanically flawless:** all four midpoints AND all four side lengths ($EF=\sqrt{37}/2$, $FG=4$, $GH=\sqrt{37}/2$, $HE=4$) computed correctly — the only gap was writing the final one-line conclusion (see Areas to Revise).
- Number-line plotting done properly this time:** Q3(iv)(b)'s solution points were confirmed marked with circles at both -2 and 7 on deep-check — this was not the gap it first appeared to be.
- MCQ consistency:** 11/12, in line with Seemab's established baseline, with only the "reference position" terminology tripping her up.

Areas to Revise

- Final-line transcription care:** three of this exam's losses are "working correct, final line wrong or missing": the dropped x^2 in Q2(vii) (a), the stray exponent left on Q2(ix)'s root, and the unwritten quadrilateral classification in Q3(v)(c). In every case the mathematics needed for full marks was already sitting on the page — the habit to build is double-checking the very last line written before moving on.
- Inequality-direction consistency when a case flips the sign:** Q3(iv)(a)'s case 2 was written with one inequality direction, then silently switched to the other with no working shown for why. The final answer happened to be correct, but every direction change needs its own justification line (e.g. "dividing/multiplying by a negative reverses the sign").
- Digit-level arithmetic care in logarithm characteristic calculations:** 40×0.8451 was computed as 34.04 instead of 33.804 in Q3(ii) (b), flipping the final answer from 34 digits to 35. A quick estimate check ($40\times 0.85 \approx 34$, so the answer should be just under 34, not over) would have caught this.
- Careful reading of large numbers in word problems:** Rs. 600,000 became Rs. 6,000,000 in Q3(i)(b), a $10\times$ slip that cost 1.5 marks despite a completely correct ratio method. Drill: underline or circle every given numeric value before starting the calculation.
- Recurring pattern — confirmed 3rd occurrence:** the missing final naming/classifying/concluding sentence has now shown up on exam-10 (missing diagonal proof), exam-12 (missing property names), and exam-13 (missing quadrilateral classification). This is the single most valuable habit to fix before the next paper.

Comparison vs previous Math attempt (exam-12-ch1-2-4-5-7, 9 Jul 2026): 62/65 (95.4%, A+, MCQ 12/12) → this attempt 58.5/65 (90.0%, A+, MCQ 11/12). The drop is concentrated in Section C (18.5/20 → 15.5/20) and one MCQ; Section B held essentially steady (32/33 both times). This also exactly matches the score and grade of exam-11's first attempt (3 Jul 2026: 58.5/65, 90.0%, A+) — the second time this exact headline result has occurred for Math.